MoFA Nepal Remittance Corridor Pipeline

Measuring the Invisible – Korea Corridor, Gulf Comparators, and EPS MOU Analysis

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Contents

1 Introduction

This report runs the full MoFA Cluster 10 remittance pipeline and embeds key outputs.

PDF prerequisites: install `tinytex` and run `tinytex::install_tinytex()`. For NRB PDF densification, install poppler (`brew install poppler` on macOS).

Render: RStudio Knit to PDF, or `rmarkdown::render("mofa_pipeline_report.Rmd")` from the project folder.

Set `params$use_real_data: false` in the YAML header to use demo data instead of collected real files.

2 Part 1 – Data preparation

2.1 RPW corridor panel

```
if (USE_REAL_DATA) {
  source_script("scripts/00b_reshape_rpw_excel.R")
} else {
  source_script("scripts/00_generate_demo_data.R")
}
```

```
##
## ================================================================
## scripts/00b_reshape_rpw_excel.R
## ================================================================
## Excel sheet names:
## [1] "Terms of Use"            "Methodology"
## [3] "Legend"                  "Countries"
## [5] "Dataset (up to Q1 2016)" "Dataset (from Q2 2016)"
##
## NOTE: This vintage uses two period-split dataset sheets (not separate $200/$500
## sheets). cc1 = USD 200 send; cc2 = USD 500 send within each sheet.
##
## Reading sheet: Dataset (up to Q1 2016) ...
##   Column names:
##  [1] "id"                       "period"
##  [3] "source_code"              "source_name"
##  [5] "source_region"            "source_income"
##  [7] "source_lending"           "source_G8G20"
##  [9] "destination_code"         "destination_name"
## [11] "destination_region"       "destination_income"
## [13] "destination_lending"      "destination_G8G20"
## [15] "firm"                     "firm_type"
## [17] "product"                  "sending location"
## [19] "speed actual"             "cc1 lcu amount"
## [21] "cc1 denomination amount" "cc1 lcu code"
## [23] "cc1 lcu fee"              "cc1 lcu fx rate"
## [25] "cc1 fx margin"            "cc1 total cost %"
## [27] "cc2 lcu amount"           "cc2 denomination amount"
## [29] "cc2 lcu code"             "cc2 lcu fee"
## [31] "cc2 lcu fx rate"          "cc2 fx margin"
## [33] "cc2 total cost %"         "inter lcu bank fx"
## [35] "transparent"              "note1"
## [37] "note2"                    "coverage"
## [39] "pick-up method"           "date"
## [41] "corridor"
```

```
##
## Reading sheet: Dataset (from Q2 2016) ...
##   Column names:
##  [1] "id"                          "period"
##  [3] "source_code"                 "source_name"
##  [5] "source_region"               "source_income"
##  [7] "source_lending"              "source_G8G20"
##  [9] "destination_code"            "destination_name"
## [11] "destination_region"          "destination_income"
## [13] "destination_lending"         "destination_G8G20"
## [15] "firm"                        "firm_type"
## [17] "payment instrument"          "access point"
## [19] "speed actual"                "cc1 lcu amount"
## [21] "cc1 denomination amount"     "cc1 lcu code"
## [23] "cc1 lcu fee"                 "cc1 lcu fx rate"
## [25] "cc1 fx margin"               "cc1 total cost %"
## [27] "cc2 lcu amount"              "cc2 denomination amount"
## [29] "cc2 lcu code"                "cc2 lcu fee"
## [31] "cc2 lcu fx rate"             "cc2 fx margin"
## [33] "cc2 total cost %"            "inter lcu bank fx"
## [35] "transparent"                 "Standard Note"
## [37] "note2"                       "receiving network coverage"
## [39] "pickup location"             "pickup method"
## [41] "date"                        "corridor"
## [43] "...43"                       "...44"
## [45] "...45"                       "...46"
## [47] "...47"
##
## Pakistan in full RPW data: YES (excluded from panel per spec)
##
## CONFIRMED: Korea -> Nepal corridor is ABSENT from RPW data.
## This gap is filled by the manually collected Korea operator fee data.
##
## Wrote 2546 rows to data/raw/rpw_real.csv
## Quarters: 54 | Corridors: 55 | Receiving countries: Bangladesh, India, Nepal, Philippines
```

```
rpw_file <- if (USE_REAL_DATA) "data/raw/rpw_real.csv" else "data/raw/rpw_demo.csv"
kable_pdf(head(read.csv(rpw_file, stringsAsFactors = FALSE), 8),
          caption = "RPW panel (first rows)")
```

2.2 Korea operator fees

```
if (USE_REAL_DATA) {
  source_script("scripts/00d_ingest_korea_fees_real.R")
}
```

```
##
## ============================================================
```

Table 1: RPW panel (first rows)

quarter	quarter_date	receiving_country	sending_country	amount_usd	total_cost_pct	mto_cost_pc
2011Q1	2011-01-01	Bangladesh	Malaysia	200	4.578	3.65
2011Q1	2011-01-01	Bangladesh	Saudi Arabia	200	4.093	4.24
2011Q1	2011-01-01	Bangladesh	Singapore	200	2.292	2.61
2011Q1	2011-01-01	Bangladesh	United Kingdom	200	5.074	5.74
2011Q1	2011-01-01	India	Canada	200	10.541	10.98
2011Q1	2011-01-01	India	France	200	7.210	7.21
2011Q1	2011-01-01	India	Germany	200	10.704	7.84
2011Q1	2011-01-01	India	Italy	200	4.171	4.50

```
## scripts/00d_ingest_korea_fees_real.R
## ============================================================
## Korea MTO fees - column names:
##  [1] "operator"            "channel_type"         "transfer_krw"
##  [4] "flat_fee_krw"        "fx_margin_pct"        "total_cost_pct"
##  [7] "collection_date"     "notes"                "cost_is_lower_bound"
## [10] "serves_nepal"        "fx_benchmark_krw_npr" "fx_benchmark_date"
##
## All rows:
##                    operator channel_type transfer_krw flat_fee_krw
##  GME / Global Money Express          MTO        5e+05         5000
##  GME / Global Money Express          MTO        1e+06         5000
##                  WireBarley          MTO        5e+05            0
##                  WireBarley          MTO        1e+06            0
##                      Sentbe          MTO        5e+05         5000
##                      Sentbe          MTO        1e+06         5000
##                       E9pay          MTO        5e+05           NA
##                       E9pay          MTO        1e+06           NA
##                     Hanpass          MTO        5e+05         5000
##                     Hanpass          MTO        1e+06         5000
##                SBI Cosmoney          MTO        5e+05         3000
##                SBI Cosmoney          MTO        1e+06         3000
##                   Rupeesend          MTO        5e+05         5000
##                   Rupeesend          MTO        1e+06         5000
##                  Korea Post         Bank        5e+05           NA
##                  Korea Post         Bank        1e+06           NA
##             KB Kookmin Bank         Bank        5e+05         8000
##             KB Kookmin Bank         Bank        1e+06         8000
##  fx_margin_pct total_cost_pct collection_date
##             NA            1.0      2026-06-24
##             NA            0.5      2026-06-24
##             NA            0.0      2026-06-24
##             NA            0.0      2026-06-24
```

```
##             NA           1.0      2026-06-24
##             NA           0.5      2026-06-24
##             NA            NA      2026-06-24
##             NA            NA      2026-06-24
##             NA           1.0      2026-06-24
##             NA           0.5      2026-06-24
##             NA           0.6      2026-06-24
##             NA           0.3      2026-06-24
##             NA           1.0      2026-06-24
##             NA           0.5      2026-06-24
##             NA            NA      2026-06-24
##             NA            NA      2026-06-24
##             NA           1.6      2026-06-24
##             NA           0.8      2026-06-24
##
##
##
##                                                       WireBarley advertises ZERO fee to Nep
##                                                       WireBarley advertises ZERO fee to Nep
##
##
##                                                                                    No pul
##                                                                                    No pul
##
##
##
##
##                                                       RupeeSend site: 5,000 KRW flat (50% s
##                                                       RupeeSend site: 5,000 KRW flat (50% s
##                                                          Korea Post Eurogiro serves only JI
##                                                          Korea Post Eurogiro serves only JI
##  Internet TT: remittance fee 3,000 (<USD5,000) + cable/telegraph fee 5,000 = ~8,000 KRW (cou
##  Internet TT: remittance fee 3,000 (<USD5,000) + cable/telegraph fee 5,000 = ~8,000 KRW (cou
##  cost_is_lower_bound serves_nepal fx_benchmark_krw_npr fx_benchmark_date
##                 TRUE         TRUE            0.0982839        2026-06-24
##                 TRUE         TRUE            0.0982839        2026-06-24
##                 TRUE         TRUE            0.0982839        2026-06-24
##                 TRUE         TRUE            0.0982839        2026-06-24
##                 TRUE         TRUE            0.0982839        2026-06-24
##                 TRUE         TRUE            0.0982839        2026-06-24
##                 TRUE         TRUE            0.0982839        2026-06-24
##                 TRUE         TRUE            0.0982839        2026-06-24
##                 TRUE         TRUE            0.0982839        2026-06-24
##                 TRUE         TRUE            0.0982839        2026-06-24
##                 TRUE         TRUE            0.0982839        2026-06-24
##                 TRUE         TRUE            0.0982839        2026-06-24
##                 TRUE         TRUE            0.0982839        2026-06-24
##                 TRUE         TRUE            0.0982839        2026-06-24
```

Table 2: Korea operator fees

operator	channel_type	transfer_krw	flat_fee_krw	fx_margin_pct	total_cost_p
GME / Global Money Express	MTO	5e+05	5000	NA	1
GME / Global Money Express	MTO	1e+06	5000	NA	0
WireBarley	MTO	5e+05	0	NA	0
WireBarley	MTO	1e+06	0	NA	0
Sentbe	MTO	5e+05	5000	NA	1
Sentbe	MTO	1e+06	5000	NA	0
E9pay	MTO	5e+05	NA	NA	N
E9pay	MTO	1e+06	NA	NA	N
Hanpass	MTO	5e+05	5000	NA	1
Hanpass	MTO	1e+06	5000	NA	0
SBI Cosmoney	MTO	5e+05	3000	NA	0
SBI Cosmoney	MTO	1e+06	3000	NA	0
Rupeesend	MTO	5e+05	5000	NA	1
Rupeesend	MTO	1e+06	5000	NA	0
Korea Post	Bank	5e+05	NA	NA	N
Korea Post	Bank	1e+06	NA	NA	N
KB Kookmin Bank	Bank	5e+05	8000	NA	1
KB Kookmin Bank	Bank	1e+06	8000	NA	0

```
##                  TRUE        FALSE            0.0982839        2026-06-24
##                  TRUE        FALSE            0.0982839        2026-06-24
##                  TRUE         TRUE            0.0982839        2026-06-24
##                  TRUE         TRUE            0.0982839        2026-06-24
##
## Wrote enriched Korea fees to data/raw/korea_mto_fees_real.csv
## Rows with cost_is_lower_bound=TRUE: 18
## Operators not serving Nepal (excluded from averages): Korea Post
```

```
if (USE_REAL_DATA && file.exists("data/raw/korea_mto_fees_real.csv")) {
  fees <- read.csv("data/raw/korea_mto_fees_real.csv", stringsAsFactors = FALSE)
  if ("notes" %in% names(fees)) fees$notes <- NULL
  kable_pdf(fees, caption = "Korea operator fees")
}
```

2.3 NRB monthly series

```
if (USE_REAL_DATA) {
  source_script("scripts/00c_ingest_nrb_real.R")
}
```

```
##
## ============================================================
## scripts/00c_ingest_nrb_real.R
```

```
## ============================================================
## NRB real data - column names:
##  [1] "month"              "bs_month"           "fiscal_year"
##  [4] "months_of_fy"       "remit_npr_bn"       "remit_ytd_npr_bn"
##  [7] "remit_ytd_usd_bn"   "fx_reserve_usd_bn" "labor_permits_new"
## [10] "data_notes"         "source"             "cal_m.x"
## [13] "cal_m.y"
##
## All rows:
##       month      bs_month fiscal_year months_of_fy remit_npr_bn
##  2022-02-01   2078 Falgun     2021/22            8           NA
##  2022-06-01   2079 Ashadh     2021/22           12           NA
##  2023-03-01  2079 Chaitra     2022/23            9           NA
##  2023-05-01          <NA>        <NA>           NA      152.615
##  2023-06-01          <NA>        <NA>           NA      189.110
##  2023-07-01          <NA>        <NA>           NA      177.410
##  2023-08-01          <NA>        <NA>           NA      150.440
##  2023-09-01          <NA>        <NA>           NA      172.695
##  2023-10-01   2080 Kartik     2023/24            4      133.820
##  2023-11-01          <NA>        <NA>           NA      183.180
##  2023-12-01    2080 Poush     2023/24            6      192.620
##  2024-01-01          <NA>     2024/25            7      198.080
##  2024-02-01          <NA>     2024/25            8      188.640
##  2024-03-01          <NA>     2024/25            9      209.750
##  2024-04-01 2081 Baishakh     2023/24           10      257.490
##  2024-05-01   2081 Jestha     2023/24           11      128.910
##  2024-06-01          <NA>        <NA>           NA      189.110
##  2024-07-01          <NA>     2024/25            1      177.410
##  2024-08-01          <NA>     2024/25            2      126.210
##  2024-09-01   2081 Ashwin     2024/25            3      144.170
##  2024-10-01          <NA>     2024/25            4      133.820
##  2024-11-01          <NA>     2024/25            5      183.180
##  2024-12-01          <NA>     2024/25            6      192.620
##  2025-01-01          <NA>     2025/26            7      198.080
##  2025-02-01          <NA>     2025/26            8      188.640
##  2025-03-01          <NA>     2025/26            9      209.750
##  2025-04-01          <NA>     2024/25           10      257.490
##  2025-05-01   2082 Jestha     2024/25           11      176.320
##  2025-06-01   2082 Ashadh     2024/25           12      189.110
##  2025-07-01  2082 Shrawan     2025/26            1      177.410
##  2025-08-01          <NA>     2025/26            2      174.670
##  2025-09-01          <NA>     2025/26            3      201.220
##  2025-10-01   2082 Kartik     2025/26            4      133.820
##  2025-11-01          <NA>     2025/26            5      183.180
##  2025-12-01          <NA>     2025/26            6      192.620
##  2026-01-01          <NA>        <NA>           NA      198.080
##  2026-02-01   2082 Falgun     2025/26            8      188.640
##  2026-03-01          <NA>        <NA>           NA      209.750
```

```
## 2026-04-01          <NA>     2025/26          10      257.490
## remit_ytd_npr_bn remit_ytd_usd_bn fx_reserve_usd_bn labor_permits_new
##           631.19             5.28              9.58                NA
##          1007.31             8.33              9.54                NA
##           903.39             6.92             10.94                NA
##               NA               NA                NA                NA
##               NA               NA                NA                NA
##               NA               NA                NA                NA
##               NA               NA                NA                NA
##               NA               NA                NA                NA
##           477.96             3.60             12.75                NA
##               NA               NA                NA                NA
##           733.22             5.52             13.69                NA
##               NA               NA             17.05                NA
##               NA               NA             17.27                NA
##               NA               NA             17.63                NA
##          1198.60             9.02             14.54                NA
##          1327.51             9.98             14.72                NA
##               NA               NA                NA                NA
##           136.93               NA                NA                NA
##           263.14               NA                NA                NA
##           407.31               NA             16.60            110654
##               NA               NA             16.70                NA
##               NA               NA             16.76                NA
##               NA               NA             16.84                NA
##          1261.01               NA             22.76                NA
##          1449.65               NA             23.08                NA
##               NA               NA                NA                NA
##               NA               NA             18.40                NA
##               NA            11.25             18.65            452324
##          1723.27            12.64             19.50            505957
##           177.41             1.27             20.03             44466
##           352.08               NA                NA                NA
##           553.31               NA             21.21                NA
##           687.13             4.88             21.52            145973
##           870.31               NA             22.13                NA
##               NA               NA             22.47                NA
##               NA               NA                NA                NA
##          1449.65            10.15             23.08            273576
##               NA               NA                NA                NA
##               NA               NA                NA                NA
##                                                            data_notes
##                        remit:cumulative_only_monthly_NA; permits:NA
##                        remit:cumulative_only_monthly_NA; permits:NA
##                        remit:cumulative_only_monthly_NA; permits:NA
##                                  seasonal_interpolation_from_anchors
##                                  seasonal_interpolation_from_anchors
##                                  seasonal_interpolation_from_anchors
```

```
##                                   seasonal_interpolation_from_anchors
##                                   seasonal_interpolation_from_anchors
##                          remit:cumulative_only_monthly_NA; permits:NA
##                                   seasonal_interpolation_from_anchors
##                          remit:cumulative_only_monthly_NA; permits:NA
##                                                         pdf_extracted
##                                                         pdf_extracted
##                                                         pdf_extracted
##                          remit:cumulative_only_monthly_NA; permits:NA
##                            remit:computed_cumulative_diff; permits:NA
##                                   seasonal_interpolation_from_anchors
##                                                         pdf_extracted
##                                                     computed_ytd_diff
##                                                     computed_ytd_diff
##                                                         pdf_extracted
##                                                         pdf_extracted
##                                                         pdf_extracted
##                                                         pdf_extracted
##                                                         pdf_extracted
##                                                         pdf_extracted
##                                                         pdf_extracted
##      remit:extracted_stated_month; permits:ft_approval_YTD_cumulative
##      remit:extracted_stated_month; permits:ft_approval_YTD_cumulative
##  remit:extracted_month1=cumulative; permits:ft_approval_YTD_cumulative
##                                                         pdf_extracted
##                                                         pdf_extracted
##      remit:extracted_stated_month; permits:ft_approval_YTD_cumulative
##                                                         pdf_extracted
##                                                         pdf_extracted
##                                   seasonal_interpolation_from_anchors
##      remit:extracted_stated_month; permits:ft_approval_YTD_cumulative
##                                   seasonal_interpolation_from_anchors
##                                                         pdf_extracted
##               source cal_m.x cal_m.y
##  original_collection      NA      NA
##  original_collection      NA      NA
##  original_collection      NA      NA
##         interpolated      NA       5
##         interpolated      NA       6
##         interpolated      NA       7
##         interpolated      NA       8
##         interpolated      NA       9
##  original_collection      NA      10
##         interpolated      NA      11
##  original_collection      NA      12
##        pdf_extracted      NA       1
##        pdf_extracted      NA       2
##        pdf_extracted      NA       3
```

```
##  original_collection      NA       4
##  original_collection       5       5
##         interpolated       6       6
##        pdf_extracted       7       7
##        pdf_extracted       8       8
##  original_collection       9       9
##        pdf_extracted      10      10
##        pdf_extracted      11      11
##        pdf_extracted      12      12
##        pdf_extracted       1       1
##        pdf_extracted       2       2
##        pdf_extracted       3       3
##        pdf_extracted       4       4
##  original_collection       5       5
##  original_collection       6       6
##  original_collection       7       7
##        pdf_extracted       8       8
##        pdf_extracted       9       9
##  original_collection      10      10
##        pdf_extracted      11      11
##        pdf_extracted      12      12
##         interpolated       1       1
##  original_collection       2       2
##         interpolated       3       3
##        pdf_extracted       4       4
##
## Column mapping (real -> expected):
##   month -> month [OK]
##   remit_npr_bn -> remit_npr_bn [OK]
##   fx_reserve_usd_bn -> fx_reserve_usd_bn [OK]
##   labor_permits_new -> labor_permits_new [OK]
##
## Attempting PDF densification for up to 80 missing months ( 132 candidate reports)...
##
##   [1] 2080-12 ... no extractable values
##   [1] 2079-06 ... no extractable values
##   [1] 2072-06 ... no extractable values
##   [1] 2072-05 ... no extractable values
##   [1] 2072-04 ... no extractable values
##   [1] 2071-12 ... no extractable values
##   [1] 2071-11 ... no extractable values
##   [1] 2071-10 ... no extractable values
##   [1] 2071-09 ... no extractable values
##   [1] 2071-08 ... no extractable values
##   [1] 2071-06 ... no extractable values
##   [1] 2071-05 ... no extractable values
##   [1] 2071-04 ... no extractable values
##   [1] 2071-03 ... no extractable values
```

```
##   [1] 2071-02 ... no extractable values
##   [1] 2071-01 ... no extractable values
##   [1] 2070-12 ... no extractable values
##   [1] 2070-11 ... no extractable values
##   [1] 2070-10 ... no extractable values
##   [1] 2070-09 ... no extractable values
##   [1] 2070-08 ... no extractable values
##   [1] 2070-06 ... no extractable values
##   [1] 2070-05 ... no extractable values
##   [1] 2070-04 ... no extractable values
##   [1] 2070-03 ... no extractable values
##   [1] 2070-02 ... no extractable values
##   [1] 2070-01 ... no extractable values
##   [1] 2069-12 ... no extractable values
##   [1] 2069-12 ... no extractable values
##   [1] 2069-11 ... no extractable values
##   [1] 2069-11 ... no extractable values
##   [1] 2069-10 ... no extractable values
##   [1] 2069-10 ... no extractable values
##   [1] 2069-09 ... no extractable values
##   [1] 2069-09 ... no extractable values
##   [1] 2069-08 ... no extractable values
##   [1] 2069-08 ... no extractable values
##   [1] 2069-06 ... no extractable values
##   [1] 2069-06 ... no extractable values
##   [1] 2069-05 ...

## FAILED: PDF parsing failure.
##   [1] 2069-05 ...

## FAILED: PDF parsing failure.
##   [1] 2069-04 ... no extractable values
##   [1] 2069-04 ... no extractable values
##   [1] 2069-03 ...

## FAILED: PDF parsing failure.
##   [1] 2069-03 ...

## FAILED: PDF parsing failure.
##   [1] 2069-02 ...

## FAILED: PDF parsing failure.
##   [1] 2069-02 ...

## FAILED: PDF parsing failure.
##   [1] 2069-01 ... no extractable values
##   [1] 2069-01 ... no extractable values
##   [1] 2068-12 ... no extractable values
##   [1] 2068-12 ... no extractable values
##   [1] 2068-11 ... no extractable values
##   [1] 2068-11 ... no extractable values
```

```
##   [1] 2068-09 ... no extractable values
##   [1] 2068-08 ... no extractable values
##   [1] 2068-06 ... no extractable values
##   [1] 2068-05 ... no extractable values
##   [1] 2068-05 ... no extractable values
##   [1] 2068-04 ... no extractable values
##   [1] 2068-03 ... no extractable values
##   [1] 2068-03 ... no extractable values
##   [1] 2068-02 ... no extractable values
##   [1] 2068-02 ... no extractable values
##   [1] 2068-01 ... no extractable values
##   [1] 2068-01 ... no extractable values
##   [1] 2067-12 ... no extractable values
##   [1] 2067-03 ... no extractable values
##   [1] 2067-02 ... no extractable values
##   [1] 2067-01 ... no extractable values
##   [1] 2023-04 ... no extractable values
##   [1] 2023-02 ... no extractable values
##   [1] 2023-01 ... no extractable values
##   [1] 2022-12 ... no extractable values
##   [1] 2022-11 ... no extractable values
##   [1] 2022-10 ... no extractable values
##   [1] 2022-09 ... no extractable values
##   [1] 2022-08 ... no extractable values
##   [1] 2022-07 ... no extractable values
##   [1] 2022-05 ... no extractable values
##   [1] 2022-04 ... no extractable values
##   [1] 2022-03 ... no extractable values
##   [1] 2022-01 ... no extractable values
##   [1] 2021-12 ... no extractable values
##   [1] 2021-11 ... no extractable values
##   [1] 2021-10 ... no extractable values
##   [1] 2021-09 ... no extractable values
##   [1] 2021-08 ... no extractable values
##   [1] 2021-07 ... no extractable values
##   [1] 2021-06 ... no extractable values
##   [1] 2021-05 ... no extractable values
##   [1] 2021-04 ... no extractable values
##   [1] 2021-03 ... no extractable values
##   [1] 2021-02 ... no extractable values
##   [1] 2021-01 ... no extractable values
##   [1] 2020-08 ... no extractable values
##   [1] 2020-07 ... no extractable values
##   [1] 2020-06 ... no extractable values
##   [1] 2020-04 ... no extractable values
##   [1] 2020-04 ... no extractable values
##   [1] 2020-03 ... no extractable values
##   [1] 2020-02 ... no extractable values
```

```
##   [1] 2020-01 ... no extractable values
##   [1] 2019-12 ... no extractable values
##   [1] 2019-11 ... no extractable values
##   [1] 2019-10 ...

## FAILED: PDF parsing failure.
##   [1] 2019-10 ...

## FAILED: PDF parsing failure.
##   [1] 2019-09 ... no extractable values
##   [1] 2019-08 ... no extractable values
##   [1] 2019-07 ... no extractable values
##   [1] 2019-06 ... no extractable values
##   [1] 2019-05 ... no extractable values
##   [1] 2019-04 ... no extractable values
##   [1] 2019-03 ... no extractable values
##   [1] 2019-02 ... no extractable values
##   [1] 2019-01 ... no extractable values
##   [1] 2018-12 ... no extractable values
##   [1] 2018-11 ... no extractable values
##   [1] 2018-10 ... no extractable values
##   [1] 2018-09 ... no extractable values
##   [1] 2018-08 ... no extractable values
##   [1] 2018-07 ... no extractable values
##   [1] 2018-06 ... no extractable values
##   [1] 2018-05 ... no extractable values
##   [1] 2018-04 ... no extractable values
##   [1] 2018-03 ... no extractable values
##   [1] 2018-02 ... no extractable values
##   [1] 2018-01 ... no extractable values
##   [1] 2017-12 ... no extractable values
##   [1] 2017-03 ... no extractable values
##   [1] 2017-02 ... no extractable values
##   [1] 2017-01 ... no extractable values
##   [1] 2011-10 ... no extractable values
##
## PDF densification summary:
##   Reports attempted: 132
##   Additional months recovered: 0
##
## Saved 39 rows to data/raw/nrb_remit_monthly_real.csv
## Date range: 2022-02 to 2026-04
## Months with remit_npr_bn: 36
```

```
if (USE_REAL_DATA && file.exists("data/raw/nrb_remit_monthly_real.csv")) {
  nrb <- read.csv("data/raw/nrb_remit_monthly_real.csv", stringsAsFactors = FALSE)
  nrb$month <- as.Date(nrb$month)
  cat("Months:", nrow(nrb), "| Range:", format(min(nrb$month), "%Y-%m"), "-",
      format(max(nrb$month), "%Y-%m"), "\n")
```

```
  cols <- c("month", "remit_npr_bn", "fx_reserve_usd_bn", "labor_permits_new", "source")
  kable_pdf(nrb[, cols, drop = FALSE])
}
```

```
## Months: 39 | Range: 2022-02 - 2026-04
```

month	remit_npr_bn	fx_reserve_usd_bn	labor_permits_new	source
2022-02-01	NA	9.58	NA	original_collection
2022-06-01	NA	9.54	NA	original_collection
2023-03-01	NA	10.94	NA	original_collection
2023-10-01	133.820	12.75	NA	original_collection
2023-12-01	192.620	13.69	NA	original_collection
2024-04-01	257.490	14.54	NA	original_collection
2024-05-01	128.910	14.72	NA	original_collection
2024-07-01	177.410	NA	NA	pdf_extracted
2024-08-01	126.210	NA	NA	pdf_extracted
2024-09-01	144.170	16.60	110654	original_collection
2024-10-01	133.820	16.70	NA	pdf_extracted
2024-11-01	183.180	16.76	NA	pdf_extracted
2024-12-01	192.620	16.84	NA	pdf_extracted
2024-01-01	198.080	17.05	NA	pdf_extracted
2024-02-01	188.640	17.27	NA	pdf_extracted
2024-03-01	209.750	17.63	NA	pdf_extracted
2025-04-01	257.490	18.40	NA	pdf_extracted
2025-05-01	176.320	18.65	452324	original_collection
2025-06-01	189.110	19.50	505957	original_collection
2025-07-01	177.410	20.03	44466	original_collection
2025-08-01	174.670	NA	NA	pdf_extracted
2025-09-01	201.220	21.21	NA	pdf_extracted
2025-10-01	133.820	21.52	145973	original_collection
2025-11-01	183.180	22.13	NA	pdf_extracted
2025-12-01	192.620	22.47	NA	pdf_extracted
2025-01-01	198.080	22.76	NA	pdf_extracted
2025-02-01	188.640	23.08	NA	pdf_extracted
2026-02-01	188.640	23.08	273576	original_collection
2025-03-01	209.750	NA	NA	pdf_extracted
2026-04-01	257.490	NA	NA	pdf_extracted
2023-05-01	152.615	NA	NA	interpolated
2023-06-01	189.110	NA	NA	interpolated
2023-07-01	177.410	NA	NA	interpolated
2023-08-01	150.440	NA	NA	interpolated
2023-09-01	172.695	NA	NA	interpolated
2023-11-01	183.180	NA	NA	interpolated
2024-06-01	189.110	NA	NA	interpolated
2026-01-01	198.080	NA	NA	interpolated
2026-03-01	209.750	NA	NA	interpolated

3 Part 2 – Load and clean RPW

```
source_script("scripts/01_load_clean_rpw.R")
```

```
##
## ==============================================================
## scripts/01_load_clean_rpw.R
## ==============================================================
## Loaded 2546 corridor-quarter observations across 55 corridors, 4 receiving countries, and 54
## Flagged 37 low-confidence corridor-quarters (num_services < 5).
## Saved cleaned panel to data/processed/rpw_panel_clean.csv
```

4 Part 3 – Gulf corridor diagnostics

```
source_script("scripts/02_corridor_diagnostics.R")
```

```
##
## ==============================================================
## scripts/02_corridor_diagnostics.R
## ==============================================================
## Latest quarter in panel: 2025-07
##
## TABLE 1 -- Nepal corridor diagnostic, latest quarter
##                       Corridor Cost $200 (%) Cost $500 (%) Fee (%)
##          Saudi Arabia -> Nepal          5.03          2.73    3.84
##         United States -> Nepal          4.41          3.14    2.90
##                 Qatar -> Nepal          4.00          2.19    3.01
##  United Arab Emirates -> Nepal          3.41          2.17    2.86
##                 India -> Nepal          3.14          1.52    2.86
##                  Oman -> Nepal          3.11          1.51    2.66
##        United Kingdom -> Nepal          2.90          1.97    1.82
##              Malaysia -> Nepal          2.76          1.97    1.52
##  FX margin (%) FX share of cost (%) Small-transfer penalty (pp)
##           1.19                 23.6                        2.30
##           1.51                 34.2                        1.26
##           0.98                 24.6                        1.81
##           0.55                 16.1                        1.24
##           0.27                  8.7                        1.61
##           0.45                 14.6                        1.60
##           1.07                 37.0                        0.93
##           1.23                 44.8                        0.79
##  # services surveyed Digital share (%)
##                   11              2730
##                   22              7730
##                    6              1670
##                   17              2940
##                   21              2380
##                    5                 0
##                   22              9090
##                   14              6430
##
## PLACEHOLDER volume-weighted average cost to Nepal (2025-07): 3.59%
```

Table 3: Nepal corridor diagnostic

Corridor	Cost..200....	Cost..500....	Fee....	FX.margin....	FX.share.of.cost....	Sm
Saudi Arabia -> Nepal	5.03	2.73	3.84	1.19	23.6	
United States -> Nepal	4.41	3.14	2.90	1.51	34.2	
Qatar -> Nepal	4.00	2.19	3.01	0.98	24.6	
United Arab Emirates -> Nepal	3.41	2.17	2.86	0.55	16.1	
India -> Nepal	3.14	1.52	2.86	0.27	8.7	
Oman -> Nepal	3.11	1.51	2.66	0.45	14.6	
United Kingdom -> Nepal	2.90	1.97	1.82	1.07	37.0	
Malaysia -> Nepal	2.76	1.97	1.52	1.23	44.8	

```
##   -> SDG 10.c target: 3.00%  |  Gap: 0.59 pp
##   -> G20/5x5 ceiling: 5.00%  |  Corridors above ceiling: 1 of 8
##
## TABLE 2 -- Cross-country comparison, latest quarter (simple corridor average)
## # A tibble: 4 x 6
##   receiving_country mean_cost_200 mean_fx_margin mean_fee mean_digital
##   <chr>                     <dbl>          <dbl>    <dbl>        <dbl>
## 1 Nepal                      3.59           0.91     2.69        4121.
## 2 Philippines                4.12           0.88     3.24        5608.
## 3 India                      5.42           1.11     4.31        5512
## 4 Bangladesh                 7.48           4.67     2.81        4522.
## # i 1 more variable: n_corridors <int>
##
## CAUTION: This is an unweighted, uncontrolled comparison -- it does not
## account for the fact that each country's corridor MIX differs (e.g. more
## Gulf-heavy vs more US/UK-heavy). See 03_panel_regression.R for the
## regression-adjusted version that controls for this via fixed effects.

## Saved output/figures/fig1_nepal_cost_trend.png

## Saved output/figures/fig2_fee_vs_fx_decomposition.png
##
## 02_corridor_diagnostics.R complete.
```

```
show_table("output/tables/table1_nepal_corridor_diagnostic.csv", "Nepal corridor diagnostic")
```

```
show_table("output/tables/table2_country_comparison.csv", "Cross-country comparison")
```

```
show_fig("output/figures/fig1_nepal_cost_trend.png")
```

Table 4: Cross-country comparison

receiving_country	mean_cost_200	mean_fx_margin	mean_fee	mean_digital	n_corridors
Nepal	3.59	0.91	2.69	4121.2	8
Philippines	4.12	0.88	3.24	5607.9	14
India	5.42	1.11	4.31	5512.0	20
Bangladesh	7.48	4.67	2.81	4521.8	11

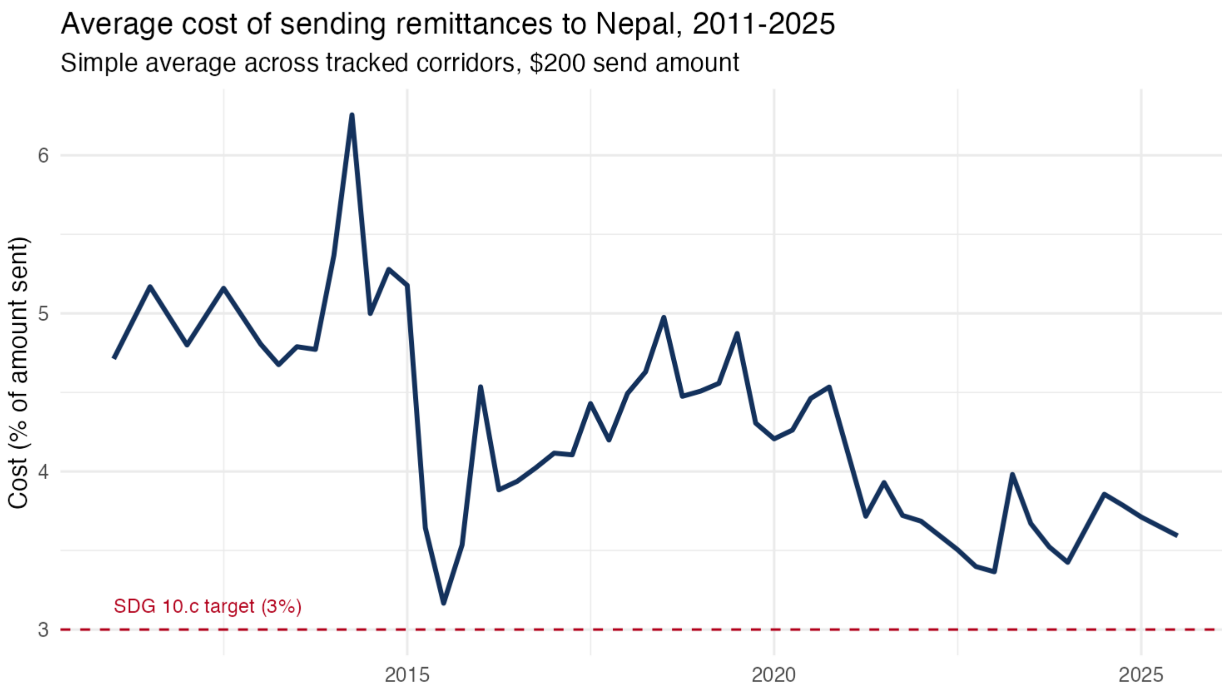

Figure 1: Nepal corridor cost trend

```
show_fig("output/figures/fig2_fee_vs_fx_decomposition.png")
```

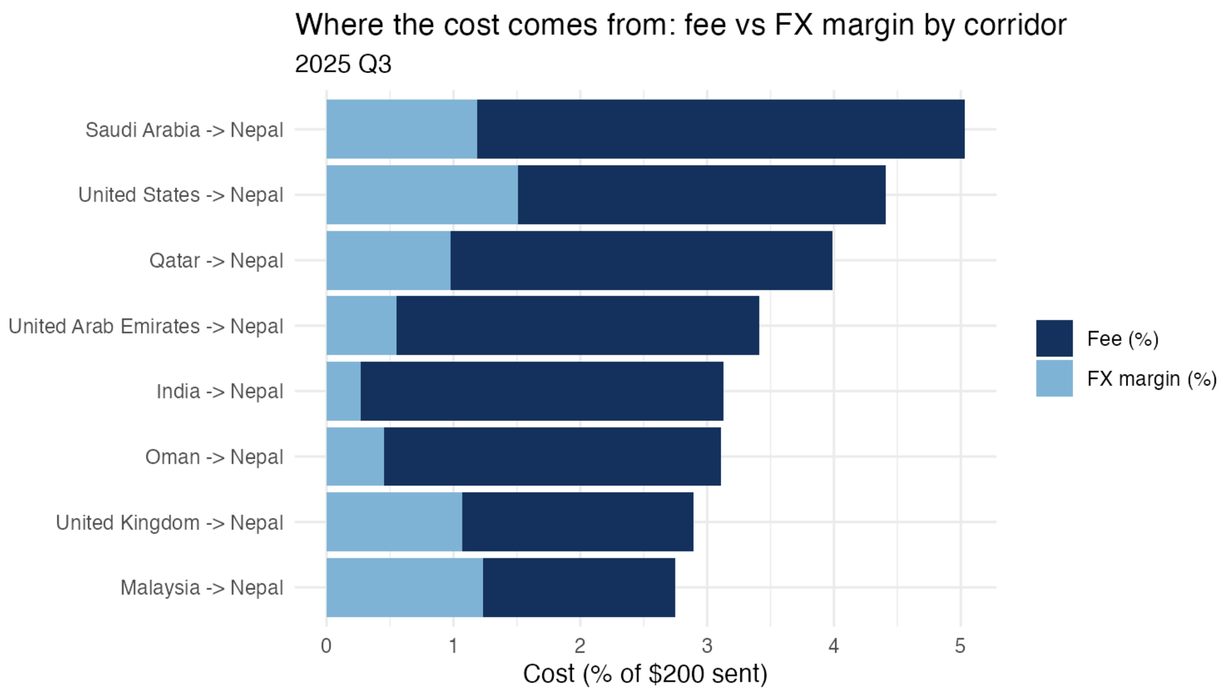

Figure 2: Fee vs FX margin decomposition

5 Part 4 – Korea cost reconstruction

```
source_script("scripts/02b_korea_cost_reconstruction.R")
```

```
##
## ============================================================
## scripts/02b_korea_cost_reconstruction.R
## ============================================================
## =================================================================
## KOREA CORRIDOR COST RECONSTRUCTION
## NOTE: fx_margin_pct is blank for 16 operators - total_cost_pct reflects
##       flat fee only (lower bound). Do not cite as final cost estimate.
## =================================================================
##
## gulf_avg_fx_margin (Nepal corridors, 2025-07): 0.908%
##
## Implied KRW/USD: 1368.5
##
## Korea → Nepal cost estimates (KRW 500,000   USD 365)
## -------------------------------------------------------
##                    operator channel_type total_cost_pct total_cost_pct_adjusted
##                  WireBarley          MTO            0.0                 0.90775
##                SBI Cosmoney          MTO            0.6                 1.50775
##  GME / Global Money Express          MTO            1.0                 1.90775
##                      Sentbe          MTO            1.0                 1.90775
```

```
##                 Hanpass          MTO            1.0                  1.90775
##               Rupeesend          MTO            1.0                  1.90775
##         KB Kookmin Bank         Bank            1.6                  2.50775
##                   E9pay          MTO             NA                       NA
##  total_cost_pct_range
##           0.00-0.91%
##           0.60-1.51%
##           1.00-1.91%
##           1.00-1.91%
##           1.00-1.91%
##           1.00-1.91%
##           1.60-2.51%
##                 <NA>
##
## MTO average lower bound (fee only):  0.77%
## MTO average upper bound (+ Gulf FX):   1.67%
##
## CROSS-CORRIDOR COMPARISON (MTO costs, apples-to-apples adjusted)
## -------------------------------------------------------
## # A tibble: 8 x 4
##   corridor          korea_lower_bound_pct korea_upper_bound_pct rpw_mto_cost_pct
##   <chr>                             <dbl>                 <dbl>            <dbl>
## 1 Korea → Nepal                     0.767                  1.67            NA
## 2 Malaysia → Nepal                NA                      NA                 2.76
## 3 Oman → Nepal                    NA                      NA                 3.11
## 4 Qatar → Nepal                   NA                      NA                 4.00
## 5 Saudi Arabia → N~               NA                      NA                 3.58
## 6 United Arab Emir~               NA                      NA                 3.32
## 7 United Kingdom →~               NA                      NA                 2.90
## 8 United States → ~               NA                      NA                 4.41
##
## *** KEY FINDING ***
## Korea formal MTO costs range from 0.77% (fee-only lower bound) to 1.67% (adding
##  the average Gulf FX margin as a conservative upper bound). Even at the upper
##  bound, Korea MTO costs remain below 7 of 7 RPW-covered Nepal corridors.

## Saved output/figures/fig_korea_operator_costs.png
##
## 02b_korea_cost_reconstruction.R complete.
```

```
show_table("output/tables/table_korea_vs_gulf_comparison.csv", "Korea vs Gulf MTO costs")

show_fig("output/figures/fig_korea_operator_costs.png")
```

Table 5: Korea vs Gulf MTO costs

corridor	korea_lower_bound_pct	korea_upper_bound_pct	rpw_mto_cost_pct
Korea → Nepal	0.77	1.67	NA
Malaysia → Nepal	NA	NA	2.76
Oman → Nepal	NA	NA	3.11
Qatar → Nepal	NA	NA	4.00
Saudi Arabia → Nepal	NA	NA	3.58
United Arab Emirates → Nepal	NA	NA	3.33
United Kingdom → Nepal	NA	NA	2.90
United States → Nepal	NA	NA	4.41

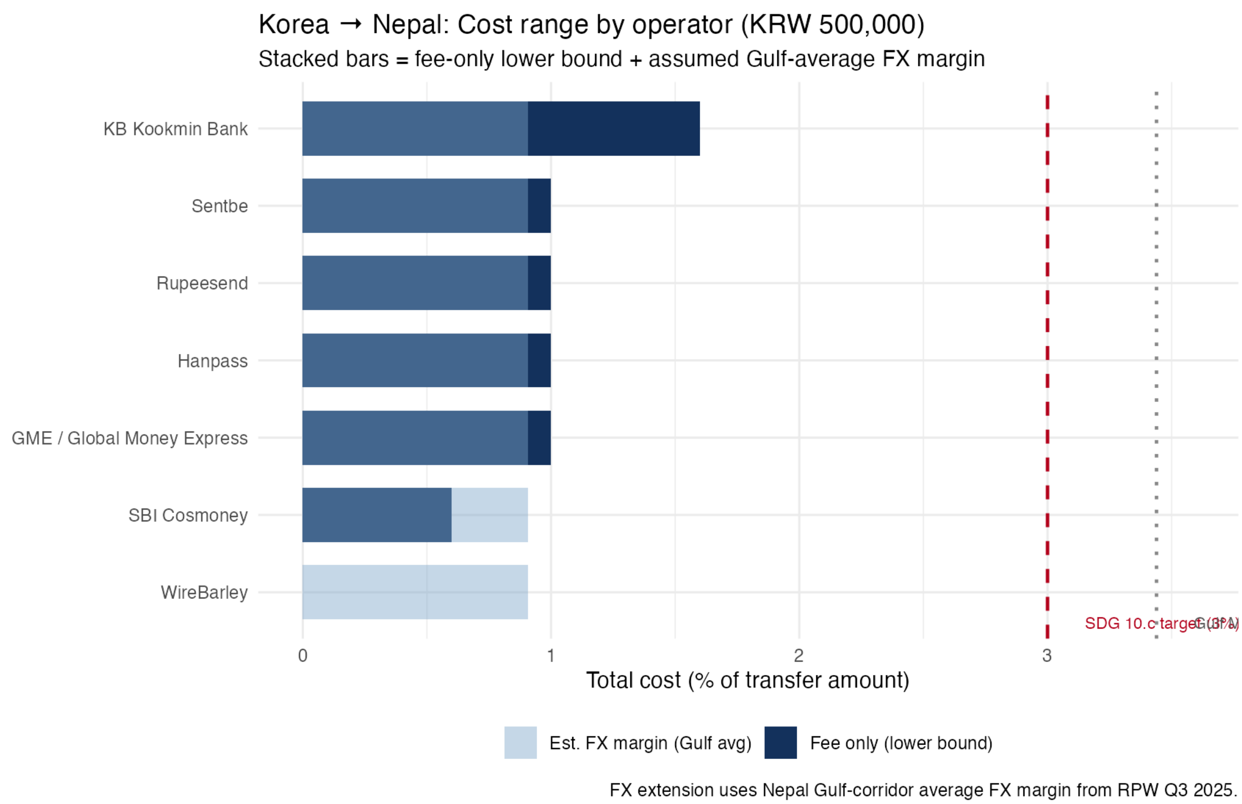

Figure 3: Korea operator costs (KRW 500,000)

6 Part 5 – Panel regression

```
source_script("scripts/03_panel_regression.R")
```

```
##
## ============================================================
## scripts/03_panel_regression.R
## ============================================================
```

```
## ================================================================
## MODEL A: Within-Nepal fixed-effects panel regression
##   total_cost_pct ~ num_services + pct_digital + corridor FE + quarter FE
## ================================================================
##
## t test of coefficients:
##
##               Estimate Std. Error t value Pr(>|t|)
## num_services 0.0155396  0.0265388  0.5855   0.5587
## pct_digital  0.0054096  0.0100800  0.5367   0.5919
##
##
## Interpretation guide:
##   - num_services coefficient: expected change in cost (pp) per additional
##     RSP surveyed in that corridor-quarter, holding corridor & time fixed.
##     A negative, significant coefficient supports the competition channel
##     from Beck & Martinez Peria (2011) -- i.e. a diplomatic case for actively
##     recruiting more licensed remittance service providers into thin corridors.
##   - pct_digital coefficient: expected change in cost (pp) per 1-unit (100pp)
##     rise in the digital share of services in that corridor-quarter. A
##     negative, significant coefficient supports prioritizing digital-payment
##     interoperability (the UPI-NPI precedent) as a cost-reduction lever.
## ================================================================
## MODEL B: Cross-country comparison, latest quarter
##   total_cost_pct ~ receiving_country + num_services + pct_digital + sending_country
##   (Nepal is the omitted/reference category for receiving_country)
## ================================================================
##
## t test of coefficients:
##
##                                    Estimate Std. Error t value  Pr(>|t|)
## (Intercept)                        1.829005   3.210282  0.5697  0.573749
## receiving_countryBangladesh        3.690090   0.453330  8.1400 1.276e-08 ***
## receiving_countryIndia            -0.711537   0.398883 -1.7838  0.086130 .
## receiving_countryPhilippines      -1.027901   0.519134 -1.9800  0.058376 .
## num_services                       0.071383   0.028701  2.4871  0.019622 *
## pct_digital                        0.013801   0.039287  0.3513  0.728197
## sending_countryBahrain            -0.147409   2.514871 -0.0586  0.953707
## sending_countryCanada             -0.566322   1.540249 -0.3677  0.716086
## sending_countryFrance              2.180707   1.488040  1.4655  0.154777
## sending_countryGermany             2.285717   1.569601  1.4562  0.157297
## sending_countryIndia              -0.520519   2.298822 -0.2264  0.822639
## sending_countryItaly               0.479963   1.525321  0.3147  0.755528
## sending_countryJapan               6.907922   2.395826  2.8833  0.007798 **
## sending_countryKuwait             -1.014156   2.086678 -0.4860  0.631028
## sending_countryMalaysia           -0.557822   1.509450 -0.3696  0.714709
## sending_countryNew Zealand         0.253850   1.449422  0.1751  0.862327
```

Table 6: Model A: within Nepal

term	estimate	std.error	statistic	p.value	model
num_services	0.02	0.03	0.59	0.56	A: within-Nepal
pct_digital	0.01	0.01	0.54	0.59	A: within-Nepal

```
## sending_countryOman                    1.404286   3.160269  0.4444  0.660462
## sending_countryQatar                   1.020700   2.497936  0.4086  0.686167
## sending_countrySaudi Arabia            2.693451   2.305077  1.1685  0.253213
## sending_countrySingapore              -1.174261   1.508150 -0.7786  0.443238
## sending_countrySouth Africa           13.830291   2.232201  6.1958 1.489e-06 ***
## sending_countrySpain                  -0.854704   1.865460 -0.4582  0.650635
## sending_countrySweden                  3.173437   1.674550  1.8951  0.069250 .
## sending_countryThailand                9.317545   1.681256  5.5420 8.088e-06 ***
## sending_countryUnited Arab Emirates  0.504392   2.230122  0.2262  0.822836
## sending_countryUnited Kingdom         -2.364972   1.790450 -1.3209  0.198049
## sending_countryUnited States          -1.040057   1.632603 -0.6371  0.529659
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
##
## Interpretation guide:
##   - Each receiving_country coefficient is that country's cost gap vs Nepal,
##     HOLDING the sending-country mix, competition, and digital share fixed.
##     Compare this to the raw, unadjusted gaps in Table 2 (script 02) -- if
##     the regression-adjusted gap is similar, your Table 2 comparison is robust;
##     if it shrinks a lot, the raw comparison was partly an artifact of corridor mix.
##
##   CAVEAT: with only ~19 corridors this is a small cross-section -- treat
##   p-values as indicative, not definitive, and say so explicitly in the brief.
##   Expanding to the FULL real RPW corridor set for these five countries
##   (likely 40-60+ corridors once you pull the real bulk file) will sharpen this.
##
## 03_panel_regression.R complete. Coefficient tables in output/tables/.
```

```
show_table("output/tables/table3a_model_within_nepal.csv", "Model A: within Nepal")
```

```
show_table("output/tables/table3b_model_cross_country.csv", "Model B: cross-country")
```

7 Part 6 – NRB time series

```
source_script("scripts/04_timeseries_structural_break.R")
```

```
##
## ============================================================
## scripts/04_timeseries_structural_break.R
## ============================================================
```

Table 7: Model B: cross-country

term	estimate	std.error	statistic	p.value	model
(Intercept)	1.83	3.21	0.57	0.57	B: cross-country
receiving_countryBangladesh	3.69	0.45	8.14	0.00	B: cross-country
receiving_countryIndia	-0.71	0.40	-1.78	0.09	B: cross-country
receiving_countryPhilippines	-1.03	0.52	-1.98	0.06	B: cross-country
num_services	0.07	0.03	2.49	0.02	B: cross-country
pct_digital	0.01	0.04	0.35	0.73	B: cross-country
sending_countryBahrain	-0.15	2.51	-0.06	0.95	B: cross-country
sending_countryCanada	-0.57	1.54	-0.37	0.72	B: cross-country
sending_countryFrance	2.18	1.49	1.47	0.15	B: cross-country
sending_countryGermany	2.29	1.57	1.46	0.16	B: cross-country
sending_countryIndia	-0.52	2.30	-0.23	0.82	B: cross-country
sending_countryItaly	0.48	1.53	0.31	0.76	B: cross-country
sending_countryJapan	6.91	2.40	2.88	0.01	B: cross-country
sending_countryKuwait	-1.01	2.09	-0.49	0.63	B: cross-country
sending_countryMalaysia	-0.56	1.51	-0.37	0.71	B: cross-country
sending_countryNew Zealand	0.25	1.45	0.18	0.86	B: cross-country
sending_countryOman	1.40	3.16	0.44	0.66	B: cross-country
sending_countryQatar	1.02	2.50	0.41	0.69	B: cross-country
sending_countrySaudi Arabia	2.69	2.31	1.17	0.25	B: cross-country
sending_countrySingapore	-1.17	1.51	-0.78	0.44	B: cross-country
sending_countrySouth Africa	13.83	2.23	6.20	0.00	B: cross-country
sending_countrySpain	-0.85	1.87	-0.46	0.65	B: cross-country
sending_countrySweden	3.17	1.67	1.90	0.07	B: cross-country
sending_countryThailand	9.32	1.68	5.54	0.00	B: cross-country
sending_countryUnited Arab Emirates	0.50	2.23	0.23	0.82	B: cross-country
sending_countryUnited Kingdom	-2.36	1.79	-1.32	0.20	B: cross-country
sending_countryUnited States	-1.04	1.63	-0.64	0.53	B: cross-country

```
## Using 36 consecutive months (2023-05 to 2026-04) for STL/ADF.
##
## ================================================================
## 1. AUGMENTED DICKEY-FULLER UNIT ROOT TEST
## ================================================================
##   Levels:           Dickey-Fuller = -2.447, p = 0.399
##   First difference: Dickey-Fuller = -3.078, p = 0.154
```

```
## Saved output/figures/fig3_stl_decomposition.png
##
## ================================================================
## 2. ENDOGENOUS STRUCTURAL BREAK DETECTION
## ================================================================
##  [1] NA NA NA NA NA NA NA NA NA NA NA NA NA NA NA NA NA NA NA NA NA NA NA NA NA
## [26] NA NA NA NA NA NA NA NA NA NA NA
```

```
## Saved output/figures/fig4_structural_breaks.png
##
## ================================================================
## 3. INTERRUPTED TIME SERIES + CHOW TEST AT NAMED EVENT DATES
## ================================================================
## Chow test failed for covid: inadmissable change point
## --- ITS model + Chow test: covid (2020-03-01) ---
##                Estimate Std. Error   t value     Pr(>|t|)
## (Intercept) 167.7456022 10.3702078 16.175722 1.548101e-17
## t             0.9165391  0.5086531  1.801894 8.043493e-02
## Chow test: skipped (inadmissible change point for this sparse series).
##
## --- ITS model + Chow test: fatf (2025-02-01) ---
##                Estimate Std. Error    t value     Pr(>|t|)
## (Intercept) 173.8407301  13.661267 12.7250815 4.600482e-14
## t             0.1361649   1.164541  0.1169258 9.076496e-01
## post         14.1233700  21.599081  0.6538875 5.178561e-01
## t_post        0.5895923   2.258981  0.2609993 7.957655e-01
## Chow test: F = 0.248, p = 0.7819
##
## --- ITS model + Chow test: upi_npi (2026-06-01) ---
## UPI-NPI break test skipped: fewer than 6 post-event observations available.
## Report qualitatively.
##
## 04_timeseries_structural_break.R complete.
```

```
show_table("output/tables/table4_chow_test_summary.csv", "Chow test summary")
```

```
show_fig("output/figures/fig3_stl_decomposition.png")
```

Table 8: Chow test summary

event	date	chow_F	chow_p	note
covid	2020-03-01	NA	NA	chow test failed
fatf	2025-02-01	0.25	0.78	
upi_npi	2026-06-01	NA	NA	skipped - only 0 post-event month(s) of data

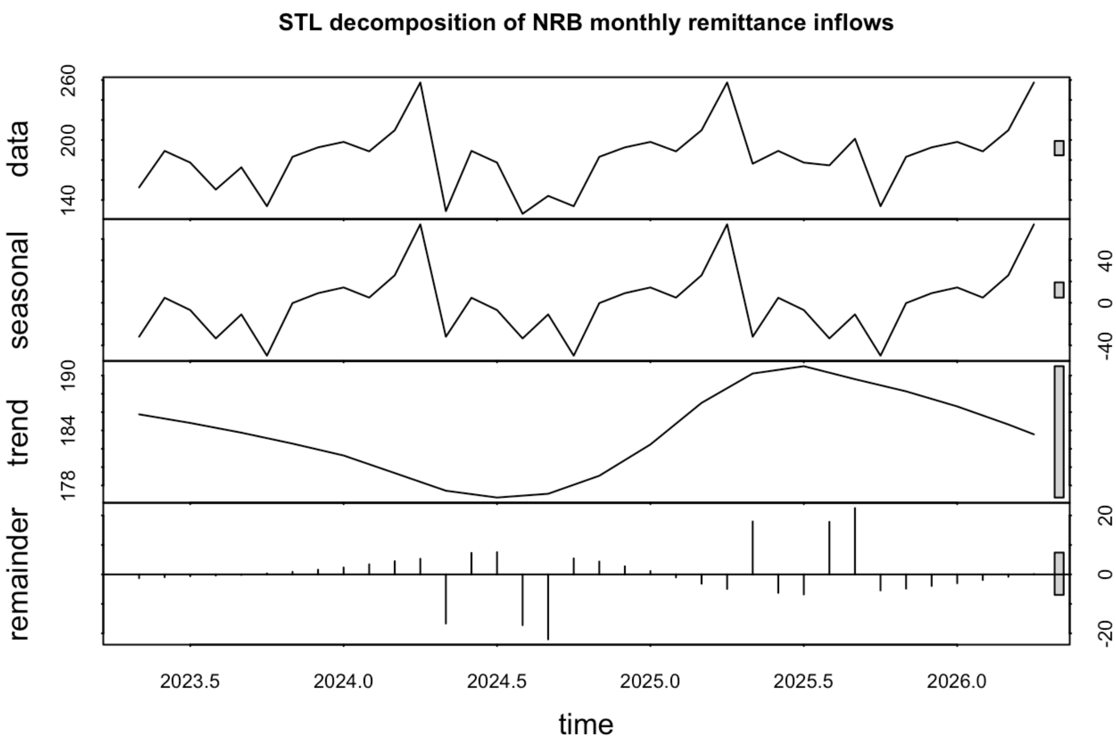

Figure 4: STL decomposition

```
show_fig("output/figures/fig4_structural_breaks.png")
```

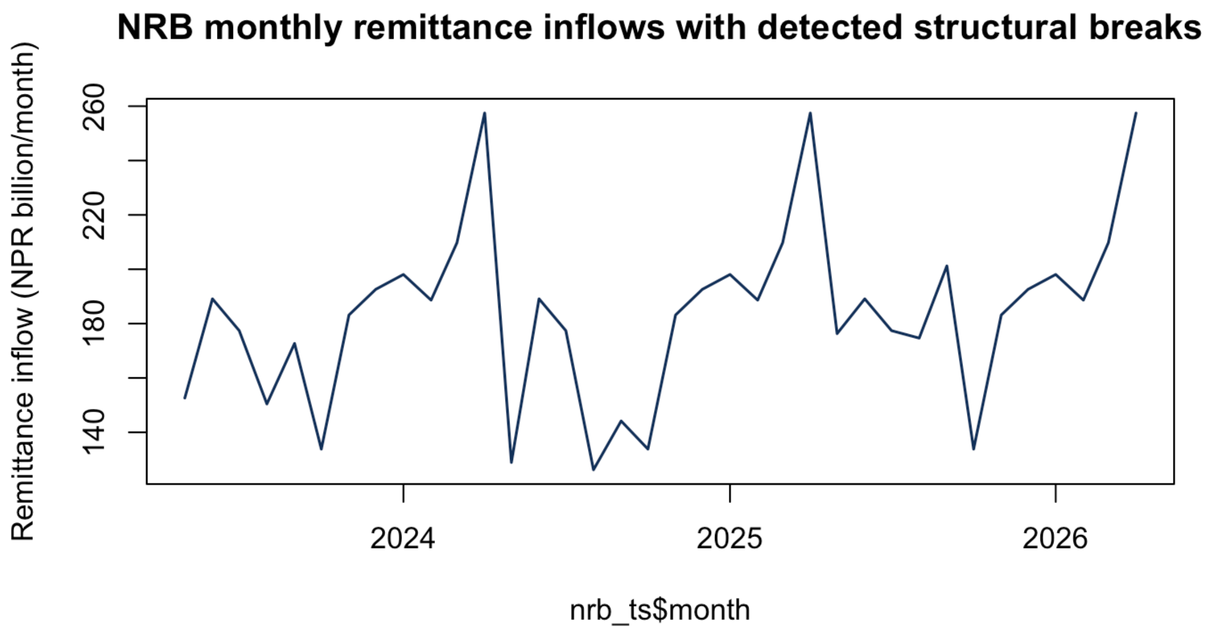

Figure 5: Structural breaks

8 Part 7 – Monte Carlo

```
source_script("scripts/05_monte_carlo_retention.R")
```

```
##
## ==============================================================
## scripts/05_monte_carlo_retention.R
## ==============================================================
## ===================================================================
## SCENARIO 1: KOREA - VALUE OF FORMALIZATION (20,000 simulations)
## ===================================================================
##   Korea corridor total: USD 484M/year (NLSS IV 2022/23)
##   Formal MTO cost: 1.30%
##   Current informal share: N(0.55, 0.10) | Target: N(0.20, 0.05)
##   Hundi implicit cost: Uniform[0.3%, 2.0%]
##
## Flows recaptured into formal/NRB-visible channels (USD M/year):
##   10th percentile: USD 99M
##   Median:          USD 168M
##   90th percentile: USD 241M
##
## Net household COST of switching [+ve = costs more; -ve = saves money]:
##   10th percentile: USD -0.9M
##   Median:          USD 0.2M
##   90th percentile: USD 1.4M
```

Table 9: Monte Carlo summary

scenario	metric	p10	median	p90	unit
S1: Korea formalization	flows_recaptured_usd_mn	99.3	168.3	240.7	USD million/year
S1: Korea formalization	net_household_cost_usd_mn	-0.9	0.2	1.4	USD million/year
S2: Saudi bank→MTO switch	household_savings_usd_mn	5.8	12.2	18.9	USD million/year

```
##
## BANK_COST_PCT: old=11.50 new=11.54 (Saudi→Nepal Q3 2025, RPW $200)
## MTO_COST_PCT:  old=3.40 new=3.58 (Saudi→Nepal Q3 2025, RPW $200)
##
## ================================================================
## SCENARIO 2: SAUDI ARABIA - VALUE OF BANK→MTO CHANNEL SWITCH (20,000 simulations)
## ================================================================
##   Saudi volume: USD 1020M/year
##   Bank cost: 11.5% | MTO cost: 3.6% (RPW Q3 2025)
##   Current bank share: N(0.18, 0.06) | Target: N(0.03, 0.02)
##
## Annual savings to Saudi-Nepal workers from channel switch (USD M/year):
##   10th percentile: USD 6M
##   Median:          USD 12M
##   90th percentile: USD 19M
##
## TABLE 5 SUMMARY (tall format):
##                    scenario                    metric  p10 median   p90
##    S1: Korea formalization   flows_recaptured_usd_mn 99.3  168.3 240.7
##    S1: Korea formalization net_household_cost_usd_mn -0.9    0.2   1.4
##  S2: Saudi bank→MTO switch  household_savings_usd_mn  5.8   12.2  18.9
##              unit
##  USD million/year
##  USD million/year
##  USD million/year
##
## Saved output/tables/table5_monte_carlo_summary.csv
```

```
## Saved output/figures/fig5_monte_carlo_distributions.png
##
## 05_monte_carlo_retention.R complete.
```

```
show_table("output/tables/table5_monte_carlo_summary.csv", "Monte Carlo summary", max_rows = 20
```

All figures in USD million/year. Scenario 1 = FX flows recaptured into formal channels if informal share falls from ~55% to ~20%. Scenario 2 = household savings if Saudi bank-channel users switch to MTOs.

```
show_fig("output/figures/fig5_monte_carlo_distributions.png")
```

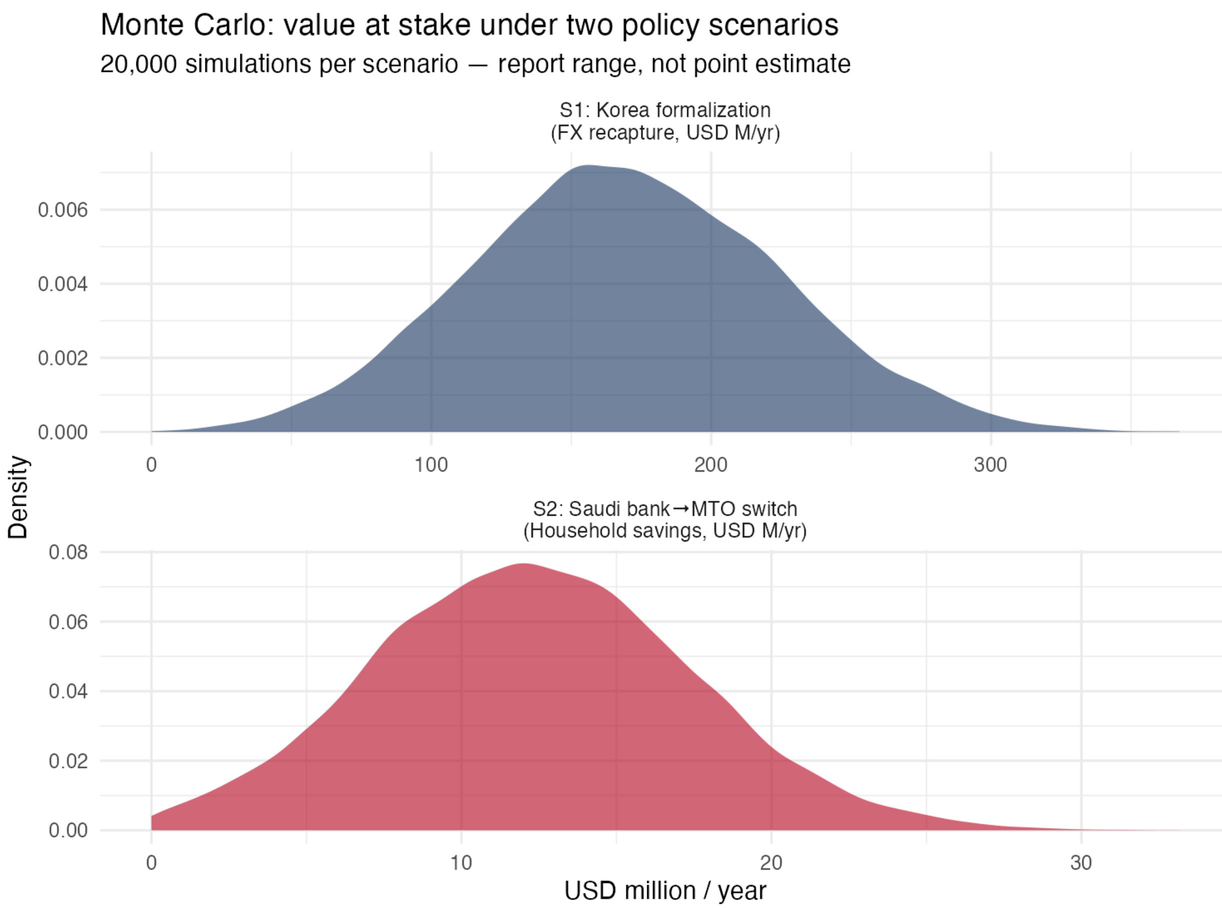

Figure 6: Monte Carlo distributions

9 Part 8 – EPS MOU analysis

```
source_script("scripts/06_eps_mou_analysis.R")
```

```
##
## ============================================================
## scripts/06_eps_mou_analysis.R
## ============================================================
## =================================================================
## EPS MOU CLAUSE ANALYSIS (2007, 21 paragraphs)
## Source: archive.ceslam.org - confirmed REAL document structure
## =================================================================
##
## Paragraphs with ANY payment/remittance provision: 0 out of 21
##
## Paragraphs with partial relevance:
##   Para 4: governs RECRUITMENT fees (sending fee) - not transfer fees
##   Para 8: pre-departure training - currently no financial/payment component
##   Para 9: labour contract - wage mentioned, payment METHOD not specified
```

```
##   Para 17: annual review - no payment agenda item exists
##   Para 20: amendment procedure - THIS IS THE DIPLOMATIC VEHICLE
##
## ================================================================
## COMPARATIVE INSTRUMENT ANALYSIS
## Which comparable frameworks have addressed payments?
## ================================================================
##
## Cross-instrument payment provision scorecard:
##
## # A tibble: 9 x 8
##   instrument                 country_pair     year has_fee_cap has_fx_disclosure
##   <chr>                      <chr>           <dbl> <lgl>       <lgl>
## 1 Nepal-Korea EPS MOU        Nepal/Korea      2007 FALSE       FALSE
## 2 Nepal-UAE MOU              Nepal/UAE        2007 FALSE       FALSE
## 3 Nepal-Malaysia MOU         Nepal/Malaysia   2018 FALSE       FALSE
## 4 Philippines-KSA POEA/BSP   Philippines/Sa~  2007 FALSE       TRUE
## 5 BSP MORB §298 (domestic)   Philippines do~  2023 FALSE       TRUE
## 6 BSP Circular 1238          Philippines do~  2026 TRUE        TRUE
## 7 NZ RSE admin arrangement   NZ/Pacific       2020 FALSE       FALSE
## 8 Korea HRD EPS worker guide Korea domestic   2023 FALSE       FALSE
## 9 UPI-NPI MOU (India-Nepal)  India/Nepal      2024 FALSE       FALSE
## # i 3 more variables: has_mto_access_clause <lgl>,
## #   has_predeparture_finlit <lgl>, has_wage_payment_method <lgl>
##
## FINDING: No bilateral labour agreement (BLA/MOU) anywhere has a
## binding fee-cap on remittance transfers. The successful instruments
## are (a) domestic central-bank regulation (BSP model) and
## (b) administrative requirements embedded in programme orientation
## (Korea HRD pre-departure guide already mentions financial literacy).
## The realistic diplomatic recommendation is therefore:
##   NOT a binding fee-cap in the EPS MOU (no precedent; unlikely)
##   BUT cooperative language + an annual-review agenda item + NRB
##   domestic regulation, using the EPS MOU Amendment (Para 20) as
##   the vehicle to add a non-binding cooperation clause.
##
## ================================================================
## PROPOSED MODEL CLAUSE LANGUAGE FOR EPS MOU AMENDMENT
## (to be reviewed by MoFA Legal Division - not a legal opinion)
## ================================================================
##
##
## PROPOSED NEW PARAGRAPH [X]: REMITTANCE AND PAYMENT CHANNEL COOPERATION
##
## (a) Recognition: The Parties recognise that the facilitation of safe,
##     affordable, and transparent remittance transfers benefits EPS workers
##     and supports the development objectives of both countries.
##
```

```
## (b) Disclosure (non-binding cooperation): The Republic of Korea agrees to
##     include, in the standard HRD Korea pre-departure education programme
##     (currently required under [relevant Korean regulation]), information on
##     licensed remittance service providers operating on the Korea-Nepal
##     corridor, including their fee schedules and exchange rate margins
##     relative to the mid-market rate at the time of transfer.
##
## (c) Monitoring: The Parties agree to exchange annually, through the
##     Joint Committee established under Paragraph 18, available data on
##     the estimated volume and cost of remittance transfers on the
##     Korea-Nepal corridor, and to jointly request the World Bank to
##     include the Korea-Nepal corridor in its Remittance Prices
##     Worldwide (RPW) monitoring database.
##
## (d) NRB-BOK cooperation: The Nepal Rastra Bank and the Bank of Korea
##     shall be encouraged to explore a cooperative framework for payment-
##     system data sharing and technical assistance, modelled on the NRB-RBI
##     Terms of Reference that supported the UPI-NPI cross-border payment
##     link (February 2024).
##
## Note: This clause is intentionally non-binding to reflect current
## precedent (no bilateral labour agreement globally contains a binding
## remittance fee-cap) while creating a structured diplomatic hook for
## progressive tightening in future MOU renewals.
## Saved output/tables/table6c_proposed_eps_clause.txt

## Saved output/figures/fig6_eps_gap_matrix.png
##
## 06_eps_mou_analysis.R complete.
```

```
show_table("output/tables/table6a_eps_mou_clause_mapping.csv", "EPS MOU clauses")
```

```
show_table("output/tables/table6b_instrument_comparison.csv", "Comparator instruments")
```

```
show_fig("output/figures/fig6_eps_gap_matrix.png")
```

Table 10: EPS MOU clauses

paragraph	subject	has_payment_provision	notes
1	Purpose and scope	FALSE	General cooperation framework
2	Definitions	FALSE	
3	Bilateral cooperation procedures	FALSE	Roster management via EPS co
4	Sending fee / recruitment cost	FALSE	Governs recruitment agency fee
5	Worker selection	FALSE	
6	Language testing	FALSE	
7	Skills testing	FALSE	
8	Pre-departure training	FALSE	Orientation only — no financial
9	Labour contract terms	FALSE	Wage listed as 'desired conditio
10	Insurance	FALSE	
11	Worker rights and protections	FALSE	General rights; no wage-paymen
12	Change of workplace	FALSE	
13	Voluntary return / re-entry	FALSE	
14	Sojourn management and repatriation	FALSE	Return airfare provision only
15	EPS computer network	FALSE	Technical HR data sharing — n
16	Dispute resolution	FALSE	
17	Coordination and review	FALSE	Annual review mechanism — bu
18	Joint Committee	FALSE	
19	Information sharing	FALSE	
20	Amendment procedure	FALSE	Mutual written consent — this
21	Entry into force / term	FALSE	2-year renewable term

Table 11: Comparator instruments

instrument	country_pair	year	has_fee_cap	has_fx_disclosure	has_mto_ac
Nepal-Korea EPS MOU	Nepal/Korea	2007	FALSE	FALSE	FALSE
Nepal-UAE MOU	Nepal/UAE	2007	FALSE	FALSE	FALSE
Nepal-Malaysia MOU	Nepal/Malaysia	2018	FALSE	FALSE	FALSE
Philippines-KSA POEA/BSP	Philippines/Saudi	2007	FALSE	TRUE	TRUE
BSP MORB §298 (domestic)	Philippines domestic	2023	FALSE	TRUE	TRUE
BSP Circular 1238	Philippines domestic	2026	TRUE	TRUE	TRUE
NZ RSE admin arrangement	NZ/Pacific	2020	FALSE	FALSE	TRUE
Korea HRD EPS worker guide	Korea domestic	2023	FALSE	FALSE	FALSE
UPI-NPI MOU (India-Nepal)	India/Nepal	2024	FALSE	FALSE	TRUE

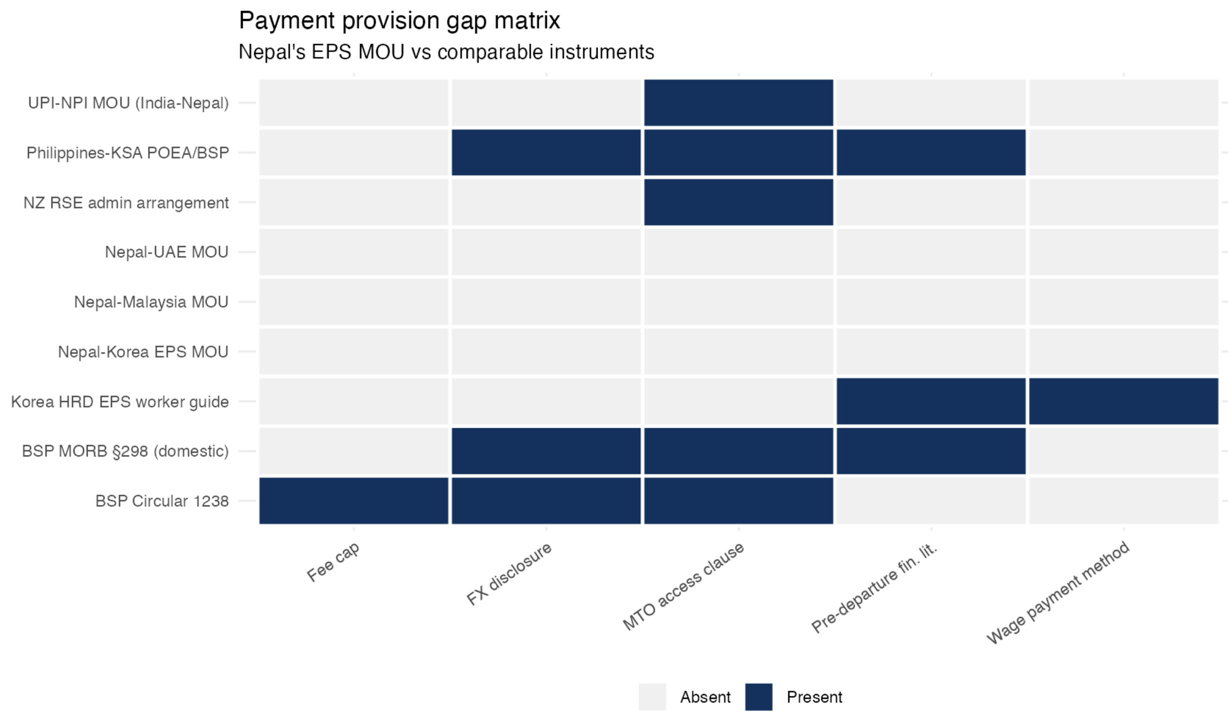

Figure 7: EPS gap matrix

10 Appendix – Proposed EPS clause

```
p <- "output/tables/table6c_proposed_eps_clause.txt"
if (file.exists(p)) cat(readLines(p), sep = "\n")
```

PROPOSED NEW PARAGRAPH [X]: REMITTANCE AND PAYMENT CHANNEL COOPERATION

(a) Recognition: The Parties recognise that the facilitation of safe, affordable, and transparent remittance transfers benefits EPS workers and supports the development objectives of both countries.

(b) Disclosure (non-binding cooperation): The Republic of Korea agrees to include, in the standard HRD Korea pre-departure education programme (currently required under [relevant Korean regulation]), information on licensed remittance service providers operating on the Korea-Nepal corridor, including their fee schedules and exchange rate margins relative to the mid-market rate at the time of transfer.

(c) Monitoring: The Parties agree to exchange annually, through the Joint Committee established under Paragraph 18, available data on the estimated volume and cost of remittance transfers on the Korea-Nepal corridor, and to jointly request the World Bank to include the Korea-Nepal corridor in its Remittance Prices Worldwide (RPW) monitoring database.

(d) NRB-BOK cooperation: The Nepal Rastra Bank and the Bank of Korea shall be encouraged to explore a cooperative framework for payment- system data sharing and technical assistance, modelled on the NRB-RBI Terms of Reference that supported the UPI-NPI cross-border

Table 12: Figures produced

file	kb
fig_korea_operator_costs.png	81.6
fig1_nepal_cost_trend.png	87.1
fig2_fee_vs_fx_decomposition.png	62.0
fig3_stl_decomposition.png	120.6
fig4_structural_breaks.png	111.3
fig5_monte_carlo_distributions.png	87.2
fig6_eps_gap_matrix.png	89.7

Table 13: Tables produced

file	kb
table_korea_mto_costs_1m.csv	2.3
table_korea_mto_costs_500k.csv	2.3
table_korea_vs_gulf_comparison.csv	0.4
table1_nepal_corridor_diagnostic.csv	0.6
table2_country_comparison.csv	0.2
table3a_model_within_nepal.csv	0.3
table3b_model_cross_country.csv	3.1
table4_chow_test_summary.csv	0.2
table5_monte_carlo_summary.csv	0.3
table6a_eps_mou_clause_mapping.csv	1.3
table6b_instrument_comparison.csv	1.0
table6c_proposed_eps_clause.txt	1.6

payment link (February 2024).

Note: This clause is intentionally non-binding to reflect current precedent (no bilateral labour agreement globally contains a binding remittance fee-cap) while creating a structured diplomatic hook for progressive tightening in future MOU renewals.

11 Appendix – Output inventory

```
inv <- function(d) {
  if (!dir.exists(d)) return(NULL)
  f <- list.files(d)
  data.frame(file = f, kb = round(file.info(file.path(d, f))$size / 1024, 1))
}
kable_pdf(inv("output/figures"), caption = "Figures produced")
```

```
kable_pdf(inv("output/tables"), caption = "Tables produced")
```